Lecture 26: Adaptive Quadrature; Eigenvalue Problems

Math 6610

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Today's Goals

- Review Gauss rules; introduce Gauss–Kronrod pairs
- Adaptive quadrature: bisect where the error estimate is large
- Eigenvalues, eigenvectors, spectrum, characteristic polynomial
- Eigenvalue algorithms must be iterative ($n \geqslant 5$)

Review

$$\int_a^b f(x)\,dx \approx G_n(f) := \sum_{i=1}^{n} w_i f(x_i).$$

Definition (*Gauss rule G_n*). Let p_n be the monic polynomial of degree n with $\int_a^b q\,p_n\,dx = 0$ for all polynomials q of degree $\leqslant n-1$. Nodes: the zeros $a < x_1 < \cdots < x_n < b$ of p_n. Weights:

$$w_i := \int_a^b \ell_i(x)\,dx, \qquad \ell_i(x) := \prod_{j\neq i} \frac{x - x_j}{x_i - x_j}.$$

Theorem (*recalled*). $G_n(p) = \int_a^b p\,dx$ for all polynomials p of degree $\leqslant 2n-1$; no n-node rule is exact for all of degree $\leqslant 2n$. Also $w_i > 0$.

Review

Nodes t_i and weights $\hat{w}_i$ on $[-1, 1]$:

n	nodes t_i	weights $\hat{w}_i$
1	0	2
2	$\pm 1/\sqrt{3}$	$1,\ 1$
3	$0,\ \pm\sqrt{3/5}$	$\frac{8}{9},\ \frac{5}{9}$ (at 0, at $\pm\sqrt{3/5}$)

Proposition (*recalled*). On $[a, b]$: $x_i := \frac{a+b}{2} + \frac{b-a}{2} t_i$, $w_i := \frac{b-a}{2} \hat{w}_i$. Exactness stays $2n - 1$.

Example ($n = 2$, $[-1, 1]$).

$$G_2(x^2) = \tfrac{1}{3} + \tfrac{1}{3} = \tfrac{2}{3} = \int_{-1}^{1} x^2 dx, \qquad G_2(x^4) = \tfrac{2}{9} \neq \tfrac{2}{5} = \int_{-1}^{1} x^4 dx.$$

Gauss–Kronrod Rules

Problem. We want an error estimate for G_n: compare it with a more accurate rule. G_{2n} is more accurate, but its nodes differ from those of G_n, so all $2n$ values of f are new.

Idea. Keep the n Gauss nodes and add $n+1$ new ones.

Definition (*Gauss–Kronrod pair*). K_{2n+1} is a rule with the n nodes of G_n plus $n+1$ new nodes in (a,b) (and its own weights). (G_n, K_{2n+1}) is a *Gauss–Kronrod pair*.

Theorem. For every n the new nodes and weights can be chosen so that $K_{2n+1}(p) = \int_a^b p\,dx$ for all polynomials p of degree $\leqslant 3n+1$.

Gauss–Kronrod Rules

rule	new values of f	exact up to degree
G_n	n	$2n-1$
G_{2n} (to check G_n)	$2n$	$4n-1$
K_{2n+1} (to check G_n)	$n+1$	$3n+1$

Remark (*Rule with error estimate*).

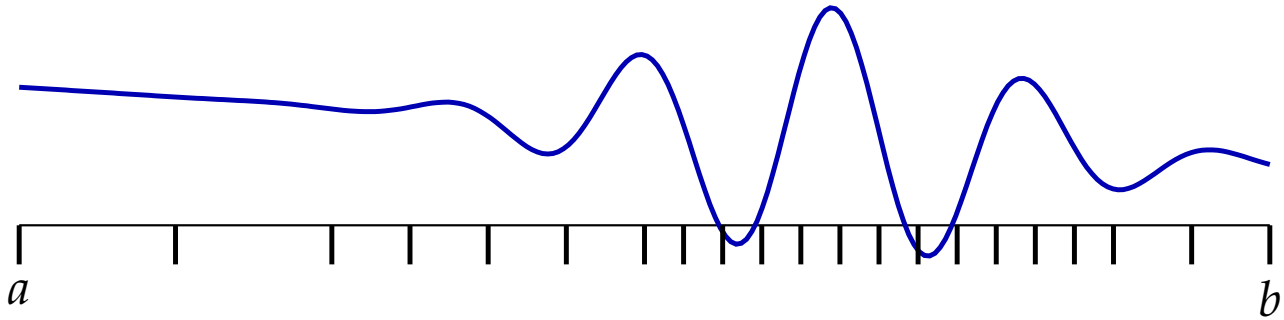
$$Q := K_{2n+1}(f), \qquad E := |G_n(f) - K_{2n+1}(f)|.$$

E estimates the error of the less accurate G_n, so it typically overestimates the error of K_{2n+1}. Matlab's `integral` uses (G_7, K_{15}).

Example ($n=1$, $[-1,1]$). $G_1(f) = 2f(0)$ and $K_3(f) = \frac{5}{9}f(-\sqrt{3/5}) + \frac{8}{9}f(0) + \frac{5}{9}f(\sqrt{3/5})$. For $f = x^2$: $G_1 = 0$, $K_3 = \frac{5}{9}\cdot\frac{3}{5}\cdot 2 = \frac{2}{3}$, and $\int_{-1}^{1} x^2 dx = \frac{2}{3}$. So $E = \frac{2}{3}$ is exactly the error of G_1.

Adaptive Quadrature

Use more values $f(x_j)$ where f varies more.



Ticks: intervals produced by the procedure ($\epsilon = 10^{-3}$, rule of the worked example).

Adaptive Quadrature

Ingredients

- A rule giving, on any interval I, $Q_I \approx \int_I f\,dx$ and an error estimate E_I (e.g. K_{2n+1} and $|G_n - K_{2n+1}|$).
- A tolerance $\epsilon > 0$.

Goal:

$$\left| Q_{[a,b]} - \int_a^b f(x)\,dx \right| \lessapprox E_{[a,b]} < \epsilon.$$

$\lessapprox$: the true error is expected to be at most about $E_{[a,b]}$ (an estimate, not a proven bound).

Adaptive Procedure

At every stage

$$[a,b] = I_1 \cup I_2 \cup \cdots \cup I_l, \qquad I_j = [a_j, b_j], \quad h_j = b_j - a_j.$$

The I_j overlap only at end points, so $h_1 + \cdots + h_l = b - a$. On each interval

$$Q_j \approx \int_{I_j} f(x)\,dx, \qquad \left| Q_j - \int_{I_j} f(x)\,dx \right| \lessapprox E_j.$$

Adaptive Procedure

Start. $l = 1$, $I_1 = [a, b]$, $h_1 = b - a$; apply Q to $I_1 \to Q_1, E_1$.

Test. If $E_j < \frac{h_j}{b-a}\,\epsilon$ for all $j = 1, \ldots, l$, stop with $Q_{[a,b]} := Q_1 + \cdots + Q_l$ and

$$E_{[a,b]} := E_1 + \cdots + E_l < \frac{\epsilon}{b-a} \sum_{j=1}^{l} h_j = \epsilon.$$

(Since $\int_a^b f = \sum_j \int_{I_j} f$, the local errors add by the triangle inequality.)

Adaptive Procedure

Bisection. Otherwise, for each j with $E_j \geqslant \frac{h_j}{b-a}\epsilon$ (right-hand sides use the old a_j, b_j):

$$I_j := \left[a_j, \frac{a_j + b_j}{2}\right], \qquad I_{l+1} := \left[\frac{a_j + b_j}{2}, b_j\right],$$

$$h_j := h_{l+1} := \frac{b_j - a_j}{2}, \qquad l := l + 1.$$

Apply Q to the two halves; intervals that passed keep Q_j, E_j. Then repeat the test.

Proportional Error Budget

I_j gets the share $\frac{h_j}{b-a}\epsilon$ of ϵ. If all pass,

$$E_{[a,b]} = \sum_j E_j < \frac{\epsilon}{b-a} \sum_j h_j = \epsilon.$$

- The share depends only on h_j: a passed interval is never re-tested.
- Equal shares ϵ/l move: l grows with every bisection, so earlier decisions would have to be redone.

Why Bisection Works

Where f is smooth and h small, suppose the local error is $\approx Ch^{p+1}$. Halving h:

$$\text{error:} \quad Ch^{p+1} \longmapsto 2^{-(p+1)}\, Ch^{p+1},$$
$$\text{budget:} \quad \tfrac{h}{b-a}\epsilon \longmapsto \tfrac{1}{2} \cdot \tfrac{h}{b-a}\epsilon.$$

So (error)/(budget) shrinks by 2^p (Simpson: $p = 4$, factor 16): a few bisections fix a smooth region. Near kinks, spikes, singularities, many bisections: work goes where f is hard.

Safeguards: $h_j \geqslant h_{\min} > 0$ and a maximum number of intervals (else x^{-1} on $(0, 1]$ or noisy f never terminates).

Worked Example

$\int_0^1 \sqrt{x}\,dx = \frac{2}{3}$, $\epsilon = 0.01$. On I: S_3 = Simpson (3 points), S_5 = Simpson on the two halves (5 points),

$$Q_I := S_5, \qquad E_I := \tfrac{16}{15}\,|S_5 - S_3|\,.$$

round	I_j	Q_j	E_j	$\frac{h_j}{b-a}\epsilon$	decision
0	$[0,1]$	0.656526	$1.97 \cdot 10^{-2}$	$1.00 \cdot 10^{-2}$	bisect
1	$[0,\frac{1}{2}]$	0.232117	$6.96 \cdot 10^{-3}$	$5.00 \cdot 10^{-3}$	bisect
1	$[\frac{1}{2},1]$	0.430962	$3.0 \cdot 10^{-5}$	$5.00 \cdot 10^{-3}$	pass
2	$[0,\frac{1}{4}]$	0.082066	$2.46 \cdot 10^{-3}$	$2.50 \cdot 10^{-3}$	pass
2	$[\frac{1}{4},\frac{1}{2}]$	0.152368	$1.1 \cdot 10^{-5}$	$2.50 \cdot 10^{-3}$	pass

(Row 0: $S_3 = \frac{1}{6}(4\sqrt{1/2}+1) = 0.638071$, $S_5 = 0.656526$, $E = \frac{16}{15}(0.018455) = 0.0197$.)

Worked Example

Round 2: $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$, $[\frac{1}{2}, 1]$ all pass.

$$Q_{[0,1]} = 0.082066 + 0.152368 + 0.430962 = 0.665396,$$
$$E_{[0,1]} = 2.46 \cdot 10^{-3} + 1.1 \cdot 10^{-5} + 3.0 \cdot 10^{-5} = 2.50 \cdot 10^{-3} < \epsilon.$$

True error: $\frac{2}{3} - 0.665396 \approx 1.27 \cdot 10^{-3} < E_{[0,1]}$.

All bisections after round 0 are next to $x = 0$, where f' is unbounded.

Eigenvalues and Eigenvectors

Definition (*Eigenvalue, eigenvector*). Let $A \in \mathbb{C}^{n\times n}$. $\lambda \in \mathbb{C}$ is an *eigenvalue* of A if there is $x \in \mathbb{C}^n$, $x \neq 0$, with

$$Ax = \lambda x.$$

Such x is an *eigenvector* of A.

($x \neq 0$ is essential: $A \cdot 0 = \lambda \cdot 0$ for every λ.)

Example: $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$: $Ae_1 = 2e_1$, $Ae_2 = 3e_2$.

The Spectrum

Definition (*Spectrum*).

$$\Lambda(A) := \left\{ \lambda \in \mathbb{C} \;\middle|\; \lambda \text{ is an eigenvalue of } A \right\}$$

Example: $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ has $\Lambda(A) = \{2, 3\}$.

The Characteristic Polynomial

Definition (*Characteristic polynomial*).

$$p_A(z) := \det(zI - A) = z^n + \alpha_{n-1}z^{n-1} + \cdots + \alpha_1 z + \alpha_0$$

Monic, degree n. Example, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$p_A(z) = (z-a)(z-d) - bc = z^2 - (a+d)\,z + (ad - bc).$$

Eigenvalues Are Zeros of p_A

Proposition. For $A \in \mathbb{C}^{n\times n}$, $\lambda \in \mathbb{C}$: $\lambda \in \Lambda(A) \Leftrightarrow p_A(\lambda) = 0$.

Proof. A square matrix B is singular exactly when $Bx = 0$ for some $x \neq 0$, and exactly when $\det B = 0$. With $B = \lambda I - A$:

$$\begin{aligned}
\lambda \in \Lambda(A) &\Leftrightarrow \exists\, x \neq 0 :\ Ax = \lambda x \\
&\Leftrightarrow \exists\, x \neq 0 :\ (\lambda I - A)x = 0 \\
&\Leftrightarrow \lambda I - A \text{ is singular} \\
&\Leftrightarrow \det(\lambda I - A) = 0 \ \Leftrightarrow\ p_A(\lambda) = 0.
\end{aligned}$$

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How Many Eigenvalues?

Remark (*Consequence*). By the fundamental theorem of algebra,

$$p_A(z) = (z - \lambda_1) \cdots (z - \lambda_n), \qquad \lambda_i \in \mathbb{C}.$$

A has n eigenvalues counted with multiplicity (not necessarily distinct).

Remark (*Two directions*).

- eigenvalues $\to$ roots: $\Lambda(A)$ = zeros of $\det(zI - A)$.
- roots $\to$ eigenvalues: every monic p of degree n equals p_C for a *companion matrix* $C \in \mathbb{C}^{n \times n}$ (next lecture).

Iterative Algorithms

Theorem (*Abel–Ruffini*). For every $n \geqslant 5$ there is a degree-n polynomial with integer coefficients whose roots are not expressible by $+$, $-$, $\times$, $\div$ and kth roots of the coefficients.

Proposition. For $n \geqslant 5$, no algorithm using finitely many arithmetic operations and kth roots computes the eigenvalues of every $A \in \mathbb{C}^{n \times n}$.

Proof. Otherwise apply it to the companion matrix C of the polynomial p above (entries 0, 1, and negatives of the coefficients of p): it would express every root of p in these operations, contradicting the theorem. ∎

So for $n \geqslant 5$ every method is *iterative*: it produces approximations converging to the eigenvalues.

Polynomials Are a Poor Route

Remark (*Why not use p_A?*).

- Roots can be extremely sensitive to coefficients (Wilkinson's polynomial $\prod_{k=1}^{20}(x-k)$).
- Matlab's `roots` goes the other way: companion matrix, then eigenvalues.

A 2×2 Warm-Up

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}: \qquad p_A(z) = (z-2)^2 - 1 = (z-1)(z-3).$$

$\Lambda(A) = \{1, 3\}$, with

$$A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Eigenvectors are orthogonal (A symmetric). Check: $\operatorname{tr} A = 4 = 1 + 3$, $\det A = 3 = 1 \cdot 3$ (sum $=$ trace, product $=$ determinant).

Sensitivity

Perturb one entry by 10^{-10}:

$$\hat{A} = \begin{bmatrix} 2 & 1 \\ 1 + 10^{-10} & 2 \end{bmatrix}, \qquad p_{\hat{A}}(z) = (z-2)^2 - (1 + 10^{-10}).$$

Eigenvalues $2 \pm \sqrt{1 + 10^{-10}}$. As $\sqrt{1+t} = 1 + \frac{t}{2} + O(t^2)$,

$$2 \pm \sqrt{1 + 10^{-10}} \approx 2 \pm \left(1 + 5 \cdot 10^{-11}\right) :$$

a shift of about $5 \cdot 10^{-11}$. Benign.

Sensitivity

$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$: $p_N(z) = z^2$, double eigenvalue 0. Perturb the $(2,1)$ entry by 10^{-10}:

$$N_\delta = \begin{bmatrix} 0 & 1 \\ 10^{-10} & 0 \end{bmatrix}, \qquad p_{N_\delta}(z) = z^2 - 10^{-10}, \qquad z = \pm 10^{-5}.$$

A change of 10^{-10} moves the eigenvalues by 10^{-5}: a *square-root* response.

Remark. N has a repeated eigenvalue but only one independent eigenvector. Such fragile structure (Jordan form) is avoided by robust algorithms, which use unitary transformations (next lecture).

Looking Ahead

- **Next lecture:** Computing Eigenvalues; Unitary Similarity